

商の微分

関数 $f(x), g(x)$ が微分可能であるとき

$$1. \left\{ \frac{f(x)}{g(x)} \right\}' = \frac{f'(x)g(x) - f(x)g'(x)}{\{g(x)\}^2}$$

$$2. \left\{ \frac{1}{g(x)} \right\}' = -\frac{g'(x)}{\{g(x)\}^2}$$

(例) 次の関数を微分せよ。

$$(1) \quad y = \frac{x+2}{x^2-1}$$

$$\begin{aligned} y' &= \frac{(x+2)'(x^2-1) - (x+2)(x^2-1)'}{(x^2-1)^2} \\ &= \frac{1 \cdot (x^2-1) - (x+2) \cdot 2x}{(x^2-1)^2} \\ &= -\frac{x^2+4x+1}{(x^2-1)^2} \quad // \end{aligned}$$

$$(2) \quad y = \frac{1}{2x+1}$$

$$\begin{aligned} y' &= \frac{(1)'(2x+1) - 1 \cdot (2x+1)'}{(2x+1)^2} \\ &= -\frac{2}{(2x+1)^2} \quad // \end{aligned}$$

$$(3) \quad y = \frac{1 - \sin x}{1 + \cos x}$$

$$\begin{aligned} y' &= \frac{(1 - \sin x)'(1 + \cos x) - (1 - \sin x)(1 + \cos x)'}{(1 + \cos x)^2} \\ &= \frac{-\cos x(1 + \cos x) - (1 - \sin x)(-\sin x)}{(1 + \cos x)^2} \\ &= \frac{\sin x - \cos x - 1}{(1 + \cos x)^2} \end{aligned}$$

$$(4) \quad y = \frac{1}{\cos x}$$

$$\begin{aligned} y' &= \frac{(1)'(\cos x) - 1 \cdot (\cos x)'}{\cos^2 x} \\ &= \frac{\sin x}{\cos^2 x} \end{aligned}$$

$$(5) \quad y = \frac{e^x}{x}$$

$$\begin{aligned} y' &= \frac{(e^x)'x - e^x(x)'}{x^2} \\ &= \frac{e^x \cdot x - e^x \cdot 1}{x^2} \\ &= \frac{e^x(x-1)}{x^2} \quad // \end{aligned}$$

$$(6) \quad y = \frac{\log x}{x}$$

$$\begin{aligned} y' &= \frac{(\log x)'x - (\log x)(x)'}{x^2} \\ &= \frac{\frac{1}{x} \cdot x - (\log x) \cdot 1}{x^2} \\ &= \frac{1 - \log x}{x^2} \quad // \end{aligned}$$